

Subject: RE: Methods in Ecology and Evolution - Decision on Manuscript ID MEE-26-04-325
Date: Monday, 29 June 2026 at 12.36.27 Central European Summer Time
From: MEE
To: Owen R. Jones
CC: esimmon2@ed.ac.uk
Attachments: image001.jpg

Hi Owen,

Hope you are well! That's no problem- I have extended the deadline to the 5th September.

Hope you enjoy your time off!

Best wishes,
Harriet

Harriet Rhodes (she/her)
Senior Assistant Editor - *Methods in Ecology and Evolution*
British Ecological Society

The British Ecological Society runs a four-day working week to support a better work/life balance. My normal working days are Monday to Thursday, with Friday as a non-working day.

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From: Owen R. Jones <jones@biology.sdu.dk>
Sent: 28 June 2026 16:28
To: MEE <coordinator@methodsinecologyandevolution.org>
Cc: esimmon2@ed.ac.uk
Subject: Re: Methods in Ecology and Evolution - Decision on Manuscript ID MEE-26-04-325

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Dear Methods in Ecology and Evolution team,

Thank you for the decision on our manuscript and for the constructive comments from the reviewers. We're pleased to see that the revisions are achievable and are now keen to prepare a careful response.

The current deadline for the revised manuscript is 10 July. However, the Danish holiday period is just beginning, and I will be on vacation until around 10 August. Would it be possible to extend the revision deadline to 5 September?

I realise this is a relatively long extension, but it would be very helpful and would allow us to address the comments properly!

Best wishes,
Owen

From: Methods in Ecology and Evolution <onbehalf@manuscriptcentral.com>
Date: Saturday, 30 May 2026 at 04.56
To: Owen R. Jones <jones@biology.sdu.dk>
Cc: Owen R. Jones <jones@biology.sdu.dk>; esimmon2@ed.ac.uk
<esimmon2@ed.ac.uk>
Subject: Methods in Ecology and Evolution - Decision on Manuscript ID MEE-26-04-325

29-May-2026

MEE-26-04-325 When does sampling uncertainty matter in matrix population models? Evidence from published projection matrices

Dear Dr Owen Jones,

Thank you for submitting your manuscript to Methods in Ecology and Evolution. I have now received the reviewers' reports and a recommendation from the Associate Editor who handled the review process. Copies of their reports are included below. This manuscript has the potential to make a valuable contribution to the area, but there are a number of significant concerns that need to be addressed. I have considered your paper in light of the comments received and I would like to invite you to prepare a major revision.

In your revision, please make sure that you take full account of the above comments and those made in the reports below. Please note that Methods in Ecology and Evolution does not automatically accept papers after revision, and an invitation to revise a manuscript does not represent commitment to eventual publication on our part. We will reject revised manuscripts if they are returned without satisfactory responses to the reviewers' comments. When returning the revised paper, please show point-by-point how you have dealt with the various comments in the appropriate section of the submission form.

We are also offering language editing with Writefull, free of charge, which may be of interest to

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Please return your revision by 10-Jul-2026. If you need longer, please let us know so we can update our system accordingly. Before resubmitting your manuscript, please read through the resubmission instructions below.

We look forward to hearing from you in due course.

Sincerely,

Huijie Qiao
Senior Editor, Methods in Ecology and Evolution

Reply to:
Ms Harriet Rhodes, Ms Amelia Macho
Methods in Ecology and Evolution Editorial Office
coordinator@methodsinecologyandevolution.org

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Associate Editor Comments to Author:
Associate Editor
Comments to the Author:

Dear Authors,

Both reviewers and I recognize the importance and originality of your work. This is an article of great relevance to comparative demographic studies, which are perhaps the most influential applications of population projection models. The results of these comparative studies are sometimes put into question because the demographic databases from which demographic metrics are originated do not provide the uncertainties of their estimates. To my knowledge, this manuscript is the first to quantify the effect of estimate uncertainty on demographic metrics and, as a result, on comparative demographic analyses.

While the article is well written, mostly clear, and it provides clearly documented code, reviewer 1 and 2 (henceforth R1 and R2) found incisive opportunities to improve the article.

Both R1 and R2 found that you should characterize the bias of your dataset in a better and more systematic way, and you should characterize its effects on your inferences. In particular, they indicated to examine taxonomic bias, and bias in the type of data that you selected (e.g. which species are we more likely to find boundary survival estimates? Which species tend to have smaller sample sizes?).

R1 has also a set of important suggestions.

First, regarding boundary survival estimates, R1 suggests to perform an analysis to explain and quantify the discrepancy between life expectancy/shape and their sampling distribution.

Second, regarding Table 1 and sampling-to-other variance ratio, R1 suggests to prominently discuss the implications of ignoring temporal variation. Shouldn't this lead to conservative estimates of sampling-to-other variance ratio?

Third, in analysis 2, R1 suggests to include a version of the point estimate model using a Poisson likelihood.

R2 also criticized your justification and choice of analyses.

First, R2 suggests to better explain and justify the relevance of analysis 1.

Second, R2 found that analysis 3 mostly replicated analysis 2. Instead, R2 suggests performing a simulation analysis (exploiting known climate coefficients) instead.

MINOR COMMENTS:

260-262: Unclear to the reader how these three options differ. What's the difference b/w pooled and mean MPM? And how do these differ from a "population-level matrix"?

351-353: This section is confusing. The equations are outside of the sentences (e.g. Eq. 11 starts after a full stop, and there is no full stop after Eq. 12), so it's hard to understand the context here.

Reviewer(s)' Comments to Author:

Reviewer: 1

Comments to the Corresponding Author

I have attached my comments.

Reviewer: 2

Comments to the Corresponding Author

Overview:

This manuscript presents an analysis of the impact of sampling uncertainty on estimates of population metrics using parametric bootstrapping methods to simulate uncertainty within published matrix projection models. The overall goal is one that would certainly be of interest to a broad audience as there is, as the authors state, a large interest in using demographic models to calculate population metrics, and perform comparative research across species. Overall, the work includes an impressive set of analyses and code that could be extremely useful as examples for other researchers to work from as propagate uncertainty in their own analyses. However, there are a few points where the manuscript needs substantial edits to make it a clear and useful methodological contribution. Primarily, the motivations for the sets of analyses should be more clearly presented as a comprehensive evaluation of the error propagation methods and their consequences, and the results could be grounded in a more biologically interesting way to make clear why these consequences or lack of consequences do or do not matter for ecologists.

While I did not specifically run all the provided code, I found the metadata to be wonderfully well documented, and the scripts that I examined seemed accurate and well-formed. The examples of different Stan models in particular seemed like would provide extremely useful examples for others to build from as they try to better account for uncertainty in their analyses.

Overall, I appreciated testing the consequences of sampling uncertainty in different contexts, across species, and within species. But there were a couple of confusing points. In Analysis 1, across species, it did not come across clearly to me as a reader why the relationship between longevity and shape of mortality is particularly interesting, or is it merely an example of a second-level analysis where you are propagating from vital rates to population matrix to population metrics to regression analysis? The authors stated that the relationship did not change materially in the regression analysis, which is encouraging, although the actual slope in figure 4B potentially did show a “substantial” change in sign.

More generally, while the authors do point out how different kinds of population metrics differ in sensitivity to uncertainty and why, it would also be interesting to do a bit of comparative demography here. Are there particular species that are more likely to have boundary-approaching survival estimates or to have limited sample sizes?

The additional value of Analysis 3 seemed limited to me, in that it mostly just replicated Analysis 2. I would be more interested to have seen an evaluation of how the importance of sampling uncertainty varied across more practical considerations like sample size, or the magnitude of the climate effect. I think this example could be framed in that way (the *Astragalus* has a weaker estimated climate signal in general than *Silene* in Analysis 2). Is that because of lower sample size or just a weaker effect, and in either case it doesn't seem to change the main takeaway that the point estimate and sampling uncertainty versions are broadly overlapping. Another suggestion would be to explore this with simulated data. I think tests with simulated data with known climate driver relationships would also help to explore how often sampling uncertainty doesn't just influence the precision of the estimates but also the accuracy.

Overall, the manuscript was clearly written and the methods seem reasonable. The main areas for improvement are more in the framing, and ensuring that the analyses robustly test why

sampling uncertainty matter in some contexts and not others.

Specific Suggestions:

Abstract:

Line 8: In the abstract, as in the introduction, it would have been valuable to more clearly define the biological questions that motivate the three analyses

Line 14: It is not clear what “It” refers to in the second sentence here because “trait” is not very clearly defined.

Line 26: The overall goal of the paper is really important and appreciated!

Introduction:

Line 40: This is somewhat personal preference, but I would argue that caring about sampling uncertainty is relevant not just in big comparative demography research, but also when working with a single species and when starting from the raw data (and that not all researchers starting from raw data do a good job at propagating uncertainty). For this reason, I would suggest re-organizing the introduction to start with the broader idea of sampling uncertainty, and then point out that it is particularly relevant to these sorts of analyses that re-use published matrices where the uncertainty is even more unavailable.

Line 55: Throughout the manuscript, I found the use of the term demographic trait to be somewhat vague. I think the text is clear enough at this line but is the trait for a population or for a species, and is it different from a life history trait.

Line 65: Understanding the consequences of sampling uncertainty is a really valuable part of this manuscript, however I do think it would be useful for this manuscript to more clearly present the methods by which the authors are incorporating this uncertainty.

Line 78: I suggest phrasing this more as “deriving parameters from MPMs that have uncertainty, and then using those parameters to testing for statistical relationships” to address question X, Y, Z. I think this would more clearly convey the chain of propagation that has to happen, and also be more clear than simply “incorporating into a statistical model”.

Line 87: How these analysis bracket this “spectrum” was not super clear to me, and also I note that there was basically no introduction of the analysis of climate drivers. The spectrum of integrated metrics to long-term expectations metrics makes sense to me, but is mostly covered by Analysis 1. A paragraph to introduce climate-driven demographic forecasting or how species responses to climate differ would be really helpful. Potentially this could also provide a bit of thinking about how uncertainty might matter in that context, and the difference between implicit year-to-year variation in matrices vs more explicit climate driver variation.

Line 95: The goal of having diagnoses seems like a very useful contribution, but what these metrics are was not clear to me after having read the full manuscript. Can there be more of a summary of this material as a table or box, somewhere in the manuscript?

Methods:

Line 122: It was not clear to me if all studies had at least 4 years of temporal replication or not?

Line 147: The authors acknowledgment of more statistically estimated vital rates is appreciated, and in general, the authors could have discussed more clearly how these results about uncertainty relate to these kinds of statistically estimated models. For example, a statistically estimated model with many more stages or ages, or even IPMS, are these likely to have less drastic consequences of uncertainty because each individual bin is sharing information across stages?

Line 168: The choice of Poisson distribution for bootstrapping reproduction is probably reasonable, but I wonder whether that fully captures the uncertainty, as often times reproduction is a highly variable. Maybe focusing on integrated reproduction (reproductives -> recruits) is overall less variable than focusing on particular uncertainty in say flower number. But something that I think deserves a bit of consideration. This also makes me wonder if there are other reproduction-centric population metrics would have different patterns compared to the focus of the current analyses on survival, population growth and life expectancy?

Line 175: It would be valuable to provide more sample sizes for the numbers of matrices for which there was no useable sample size.

Line 185: Is it the case that the sample size, and the total estimate population size is available for each of the included studies?

Line 190: It is not clear how the extra row of death frequencies is used.

Line 214: This paragraph is somewhat repetitive of paragraph at line 80, but is a more clear description of the goals of the different analyses. Moving some of this material to the introduction could be useful, and then here more quickly laying out the next analyses. This does set up Analysis 2 as part of the "Spectrum" mentioned at line 87, but I think it would still be useful to more clearly introduce why readers would care about climate-demography relationships in their own right.

Line 221: Again, I don't think the interesting aspect of robustness is really robustness across sites in Analysis 3. I suggest moving Analysis 3 to the supplemental robustness check, and instead focusing on other aspects that influence sample size and uncertainty.

Line 232: This is a nice summary of these life history metrics, but doesn't give a good sense of why it is biologically interesting that there is a relationship between shape and longevity. This is needed if this is the main focus, or the authors should more clearly test a range of different metrics.

Line 308: It is not clear what is meant by matched climate covariates if the authors need to use an external dataset (PRISM) for the climate analysis.

Line 314: It is confusing that the authors state that the only focus on reproduction, when they also analyze survival.

Line 319: This sentence here is a bit odd in this location in that it references the results that haven't appeared yet (that there is no sensitivity to sampling uncertainty). But a valuable point to make in the manuscript.

Line 350: In the description of the climate analysis, it would be helpful to more clearly sign post that the different models are the point-estimate vs the sampling uncertainty. The authors mention that there is a structural difference in addition to the sampling uncertainty component, but it could be more clear that they are estimating statistical models for both cases, and what is different is that there is a some type of measurement error model that accounts for variation in that metric in the second case.

Line 364: Additionally, this survival analysis highlights that it would be valuable to describe and compare different methods. If you only have a survival proportion in these large compilation studies, that is very different from having individual level data. You could fit the model $\text{surv} \sim \text{tmp}$ with individual level data as a Bernoulli process, or you could fit a binomial distribution with the $N_{\text{success}}/N_{\text{trials}}$ and fully propagate uncertainty in both cases. It would be helpful to readers to know what about these methods is actually doing the uncertainty propagation.

Line 372: Again, I think it is valuable to compare this analysis across two different species, and also across populations, but I still think this could be combined with Analysis 2 or moved to the supplement as the results are essentially identical.

Results:

Line 391: I suggest focusing first on the comparison across all the metrics that vary across the spectrum presented by the authors, and then move onto these "parameters of interest". Also noting that here, the authors use parameter instead of trait or metric.

Line 403: Authors state that the uncertainty only moderately influenced the result, but the actual figure shows a sign flip (although in both cases overlapping zero. So I don't know that it is clearly a moderate influence.

Line 409: I believe the authors explanation here about posterior widths, but it does not come across clearly in Figure 4 that the posterior uncertainty of the relationship is meaningful different.

Line 426: recommend saying "the posterior interval was 10.9% wider". And also it was not clear which Beta estimate is referred to here. Figure 5B? In the figure, the point estimates of survival-climate parameters in 5C is wider for the point estimate, rather than the other way around. And then at the end of this paragraph, the authors justify that the uncertainty model will have more precise estimates, rather than wider?

Discussion:

Line 453: it is not clear what is meant by trait dependent. In this case, I would also start the discussion focused on first the results of analysis 1, and then move to analysis 2 later. The two analyses are quite different so it seems unjustified to compare them too directly.

Line 495: I really appreciated this takeaways sections. I did wonder how longevity analysis of

age-based models would differ. And from analysis that are built directly from the data. I think at least presenting these as options and how they are different, because they are common use cases for demographic models, would be useful.

Figures:

Across figures, the points (representing point estimates) and the sampling uncertainty distributions all appeared in green, but the captions often mentioned purple. Showing these values in different colors would be really useful because it would highlight that the points are not a posterior mean or something directly connected to the sampling distribution. This was particularly relevant in Figure 3 and 4, where I was at first confused how you could possibly have a central tendency outside of the posterior distribution.

Figure 4: Please state what the main takeaway from the figure is in captions

Figure S1 and S2 seem like they are showing the same thing, and I think S2 is actually more useful. Also I'm curious if these boundary effects go away if estimating more continuous stages, and is this a relict of some arbitrary cutoff.

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REVISION SUBMISSION INSTRUCTIONS

Please note that resubmitting your manuscript does not guarantee eventual acceptance, and that your revision may be subject to re-review before a decision is rendered. Please also ensure that your altered manuscript still conforms to our word limit of 7000-8000 words for Research Articles, or 3000-4000 for Applications and Practical Tools articles.

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